Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/19/2026
PI Name:	Adler, Robert M.D.		
Project Title:	Magnetic Resonance Imaging (MRI) to Assess Changes to Trabecular Microarchitecture of the Hip		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jean.sibonga-1@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	804-675-5424
Organization Name:	Richmond Veterans Affairs Medical Center		
PI Address 1:	Department of Veterans Affairs (VA)		
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City:	Richmond	**State:**	VA
Zip Code:	23249-0001	**Congressional District:**	4
Comments:	Contact Co-PI email for communications.		
Project Type:	Ground	**Solicitation / Funding Source:**	2021 HERO NNJ21ZSA001N-FLAGSHIP, Appendix A: NASA Research and Technology Development to Support Crew Health and Performance in Space Exploration Missions
Start Date:	12/01/2022	**End Date:**	11/30/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2026 per PI (Ed., 12/5/24).		
Key Personnel Changes/Previous PI:	Ed. Note: The name of Dr. Adler's institution has changed; the former McGuire Research Institute, Inc. is now known as the Richmond Veterans Affairs Medical Center. Update: Chamith Rajapakse, Ph.D. (University of Pennsylvania) was added to the project in May, 2023. Dr. Rajapakse is a subject matter expert in magnetic resonance imaging (MRI) and HR-pQCT. Dimitri Martel, Ph.D. (New York University Langone) has also been added to the investigation; Dr. Martel performs the analysis of the hip scans by MRI. (Ed., 9/30/23)		
COI Name (Institution):	Sibonga, Jean Ph.D. (NASA Johnson Space Center) Gorgey, Ashraf Ph.D. (Richmond Veterans Affairs Medical Center) Chang, Gregory M.D. (New York University Langone) Young, Millennia Ph.D. (NASA Johnson Space Center) Rajapakse, Chamith Ph.D. (University of Pennsylvania) Martel, Dimitri Ph.D. (New York University Langone)		
Grant/Contract No.:	80NSSC23K0382		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	New Task Description (Ed., 12/01/24) Background. Studies of bone loss in ISS astronauts, using quantitative computed tomography (QCT) of the hip and spine and high-resolution-peripheral QCT (HRpQCT) of the lower leg, describe losses of trabecular bone mineral density (Trb vBMD) that are not detectable by the clinical test for age-related osteoporosis (Op), i.e., areal bone mineral density (aBMD) measured by dual-energy X-ray absorptiometry (DXA). Both QCT and HRpQCT revealed losses in Trb vBMD immediately following long duration (LD) spaceflight and continuing in some individuals even after recovery periods of a year or more. We reported rapid loss rates of spaceflight hip QCT Trb vBMD, relative to rates with terrestrial aging. In a pilot study, hip QCT showed lack of recovery of hip Trb vBMD to baseline in 4 of 10 astronauts evaluated 2 years after return to Earth. Analyses of serum and urine collected during spaceflight showed significant in-flight elevation of bone resorption (BR) markers in the 4 astronauts who did not recover Trb vBMD vs. the 6 astronauts who did; a biomarker for bone formation (BF) remained stable until ~120 days into the flight. Biochemical assays consistently show BR exceeding BF during spaceflight accounting for a rapid loss of Trb vBMD which could also increase the risk for disrupted trabecular connectivity. While the efficacy of an oral bisphosphonate to mitigate spaceflight bone loss has been substantiated in an astronaut flight study, there is reduced willingness in some stakeholders to use a pharmaceutical countermeasure. Hence, there is a concern for the risk of irreversible losses in trabecular connectivity in the absence of an anti-resorptive countermeasure. Such trabecular disruptions are associated with skeletal fragility and fractures in other populations undergoing bone loss but have not been characterized in astronaut hips because of prohibitive radiation exposures needed for high-resolution imaging of deeply embedded bone. HRpQCT has low radiation and sufficient resolution to study microarchitecture, but only for peripheral sites such as the lower leg and wrist. Hence, it is critical to apply a technology to the deeply embedded hip bone with no ionizing radiation, such as MRI, to safely define the effect size of rapid bone loss during spaceflight on trabecular microarchitecture, particularly with the reported observations of delayed or no recovery in hip Trb vBMD. Hypothesis. We hypothesize that 1) LD astronauts will have an increased risk for bone fractures because of disruptions in Trb uArch of the hip and 2) a 3-Tesla (3T) MRI scanning protocol can detect disruptive changes in Trb uArch, some of which could be irreversible. Specific Aims: We will investigate this hypothesis by addressing the following Specific Aims (SA): SA 1) Use spinal cord injury (SCI) to model the skeletal effects of LD spaceflight on hip trabecular bone microarchitecture (Trb uArch). SA 2) Apply a previously validated 3T MRI protocol to assess serial changes in hip Trb uArch (over a 12-mth time period) following spinal cord injury (SCI) in consented subjects. SA 3) Verify that 3T MRI data can discriminate disrupted hip Trb connectivity in subjects with SCI relative to age- and sex-matched ambulatory controls. SA 4) Validate that an 3T MRI protocol can detect disruptions in the connectivity of Trb bone uArch in the distal tibia (ankle region) in subjects with SCI as compared to detection by HRpQCT. SA 5) Compare estimates of trabecular bone strength (as a surrogate for fracture risk) at hip and the ankle by Finite Element modeling of Trb uArch data, including verification and validation processes informed by data. Data of Trb uArch are derived from serial testing (12-mth apart) of hip and ankle in study subjects (able-bodied controls and SCI) by 3T MRI and HRpQCT. ---------- New Task Description (Ed., 9/30/23) Background. Studies of bone loss in ISS astronauts, using quantitative computed tomography (QCT) of the hip and spine and high-resolution-peripheral QCT (HRpQCT) of the lower leg, describe losses of trabecular bone mineral density (Trb vBMD) that are not detectable by the clinical test for age-related osteoporosis (Op), i.e., areal bone mineral density (aBMD) measured by dual-energy X-ray absorptiometry (DXA). Both QCT and HRpQCT revealed losses in Trb vBMD immediately following long duration (LD) spaceflight and continuing in some individuals even after recovery periods of a year or more. We reported rapid rates of spaceflight hip QCT Trb vBMD loss, relative to terrestrial aging rates. In a pilot study, hip QCT showed lack of recovery of hip Trb vBMD to baseline in 4 of 10 astronauts evaluated 2 years after return to Earth. Analyses of serum and urine collected during spaceflight showed significant in-flight elevation of bone resorption (BR) markers in the 4 astronauts who did not recover Trb vBMD vs. the 6 astronauts who did; a biomarker for bone formation (BF) remained stable until ~120 days into the flight. Biochemical assays consistently show BR exceeding BF during spaceflight accounting for a rapid loss of Trb vBMD which could also increase the risk for disrupted trabecular connectivity. While the efficacy of an oral bisphosphonate to mitigate spaceflight bone loss has been substantiated in an astronaut flight study, there is reduced willingness in some stakeholders to use a pharmaceutical countermeasure. Hence, there is a concern for the risk of irreversible losses in trabecular connectivity in the absence of an anti-resorptive countermeasure. Such trabecular disruptions are associated with skeletal fragility and fractures in other populations undergoing bone loss but have not been characterized in astronaut hips because of prohibitive radiation exposures needed for high-resolution imaging of deeply embedded bone. HRpQCT has low radiation and sufficient resolution to study microarchitecture, but only for peripheral sites such as the lower leg and wrist. Hence, it is critical to apply a technology with no ionizing radiation such as MRI to safely define the effect size of spaceflight on hip trabecular microarchitecture, particularly with the observation of delayed or no recovery in hip Trb vBMD. Hypothesis. We hypothesize that 1) LD astronauts will have an increased risk for bone fractures because of disruptions in Trb uArch

	of the hip and 2) a 3-Tesla (3T) MRI scanning protocol can detect disruptive changes in Trb uArch, some of which could be irreversible. Specific Aims: We will investigate this hypothesis by addressing the following Specific Aims (SA): Specific Aims: 1) Identify and consent individuals with SCI to assess the effects of prolonged unloading to the hip; 2) Apply a previously-validated 3T MRI protocol to characterize serial changes in hip trabecular microarchitecture of SCI subjects at 0 and 12-mo following injury; 3) Compare the ability of 3T MRI to discriminate loss of trabecular connectivity between subjects with SCI (Aim 1) and age- and sex-matched ambulatory controls for the hip and ankle; 4) Evaluate the ability of MRI to discriminate changes in trabecular connectivity at the hip and ankle (by MRI serial testing 12 months apart) and the subsequent effects on trabecular bone strength (by the analysis of Finite Element models of serially acquired trabecular microarchitecture data).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Disruptions in trabecular bone microarchitecture is part of the definition of Osteoporosis: "…a systemic skeletal disease characterized by low bone mass and microarchitectural deterioration with a consequent increase in bone fragility with susceptibility to fracture …" Am. J. Med.1991. Currently, there is no test in terrestrial medicine used to assess trabecular bone microarchitecture in deeply embedded bones. Evidence acquired would substantiate that individuals with rapid loss of aBMD at the hip, including astronauts, could be at risk for irreversible loss of trabecular connectivity and would provide compelling evidence to prescribe anti-resorptive therapy. The proposed study of an MRI protocol at New York University Langone (Langone) will assess its sensitivity to detect irreversible changes in trabecular bone microarchitecture of the deeply embedded hip bone. The protocol will be assessed, using a Siemens 3T MRI Vida (models at both Richmond Veterans Affairs Medical Center / Richmond and Langone), in the non-weightbearing hips of patients following SCI. The non-weightbearing hips of the SCI will serve as an analog for the accelerated hip bone loss that occurs in astronaut hips during spaceflight. We will consent newly admitted patients to the SCI unit who are matched in age, sex and physical characteristics of astronauts as well as to matched to able-bodied controls. The effects of twelve months hip unloading following SCI will be evaluated and compared to able-bodied controls.
Task Progress:	PROGRESS UPDATE (Ed., 12/05/24) Specific Aim 1) Use spinal cord injury (SCI) to model the skeletal effects of LD spaceflight on hip trabecular bone microarchitecture (Trb uArch). The Richmond VAMC (Veterans Affairs Medical Center) received the following study approvals enabling the recruitment, consenting and measurement of study subjects (beginning with able-bodied controls): i) approval by Institutional Review Board (IRB) 02/27/24; ii) compliance with VA Privacy Requirements 03/13/24; iii) compliance as reviewed by Information and Safety Security Officer 07/27/24 and iv) Authorization to Proceed (ATP) following collective assessment by the Richmond VA Research & Development Committee 08/08/24 with regards to scientific validity and the protection of human study subjects and Veterans Health Administration staff. • Richmond Co-PI hired 09/09/24 a new Research Administrative Assistant/Study Coordinator due to the unexpected passing of previous coordinator on 07/31/24. • 08/02/24 the NASA IRB Human Research Protection Program determined that study did not require JSC IRB review & approval after confirming that data handled by JSC team will not contain any information that would readily identify study participants, and that the NASA team will not have access to the key/cipher to link participants to their data. • the NYU MRI protocol was tested on first study subjects 08/08/24 on a Siemens 3T MRI Vida installed and operating at Richmond VAMC since Summer 2023; the NYU Co-I Chang and UPenn Co-I Rajapakse confirmed the high quality of images acquired for the hip (proximal femur) and ankle (tibia) of the first subject. Recruitment of SCI study subjects anticipated for quarter 1 of calendar year 2025. • HRP extended study performance schedule by 1 year (No Cost Extension) to accommodate the delayed Authorizations by Richmond VA committees and completion of data analysis.
Bibliography Type:	Description: (Last Updated: 01/19/2026)
Abstracts for Journals and Proceedings	Adler RA, Chang G, Rajapakse C, Gorgey A, Martel D, Sibonga JD. "Magnetic resonance imaging (MRI) to assess changes to trabecular microarchitecture of the hip." 2024 Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024